

Amazon Virtual Private Cloud (VPC)

Amazon Virtual Private Cloud (Amazon VPC) enables you to launch AWS resources into a virtual network that you've defined. It allows you to provision a logically isolated section of the Amazon Web Services (AWS) Cloud.

IP Classes

Class A: 10.10.10.1

Class B: 172.16.0.1

Class C: 192.168.0.1

Project Sizes

Small Project: 50 servers

Medium Project: 250 servers

Big Project: 1000 servers

IPv4 in VPC

You can have multiple VPCs within the CIDR block range 10.10.0.0/16 to 10.10.0.0/28.

We are choosing 10.1.0.0/24, which gives 256 IP addresses.

VPC Subnet Calculations

The below link is to calculate the VPC subnet ranges:

<https://www.davidc.net/sites/default/subnets/subnets.html>

We are choosing 10.1.0.0/24.

Number of IP addresses:

$2^{(32-24)} \rightarrow 2^8 \rightarrow 256$

Assuming an equal number of IP addresses for each subnet:

Number of IP addresses per subnet = $256 / 4 = 64$

Here is the IP address distribution for the four subnets:

Public Subnet 1 -> 10.1.0.0/26 (IP range — 10.1.0.0 to 10.1.0.63)

Public Subnet 2 -> 10.1.0.64/26 (IP range — 10.1.0.64 to 10.1.0.127)

Private Subnet 1 -> 10.1.0.128/26 (IP range — 10.1.0.128 to 10.1.0.191)

Private Subnet 2 -> 10.1.0.192/26 (IP range — 10.1.0.192 to 10.1.0.255)

Reserved IPs by AWS

AWS reserves 5 IPs in each subnet:

1. Network address
2. VPC Router

3. AWS-provided DNS
4. Reserved for future use
5. Broadcast address (not supported)

Explanation of Subnet Calculations

In networking, an IP address is a 32-bit number divided into four octets (e.g., 192.168.1.1). CIDR notation (e.g., /24, /16) determines how many bits are used for the network and how many are left for hosts.

Why do we use 32?

- An IPv4 address has 32 bits.
- CIDR notation tells how many bits are for the network.
- The remaining bits determine the number of available IPs.

Why do we use 32 - 24?

- '/24' means 24 bits are for the network, leaving 8 bits for hosts.
- The number of available IPs is calculated as:
 $2^{(32 - 24)} = 2^8 = 256$ IPs.

Why substitute values between /16 and /28?

- AWS allows VPC CIDR blocks between /16 (largest) and /28 (smallest).
- /16 provides 65,536 IPs, while /28 provides 16 IPs.

Why can't we use 32 - 31?

- If we use '/31', we get $2^{(32-31)} = 2^1 = 2$ IPs, which is not practical as AWS reserves 5 IPs.
- If we use '/32', we get $2^{(32-32)} = 2^0 = 1$ IP, which is unusable.

AWS Services Limit Per Account

Max CIDR per VPC: 5

Min size per CIDR: /28 (16 IPs)

Max size per CIDR: /16 (65536 IPs)

Amazon VPC Components

1. Subnets
2. Network Access Control List (NACL)
3. Internet Gateway
4. NAT Gateway
5. Route Tables
6. Security Groups

1. Subnets

A subnet is a range of IP addresses within a VPC.

2. Network ACL (NACL)

A NACL is an additional security layer that controls traffic in and out of subnets.

3. Internet Gateway

An Internet Gateway (IGW) allows instances with public IPs to access the internet.

4. NAT Gateway

NAT Gateway allows private subnet instances to access the internet while blocking incoming connections.

5. Route Tables

Route tables contain rules that define how network traffic is directed.

6. Security Groups

Security groups act as a firewall for EC2 instances, controlling inbound and outbound traffic.

Difference Between NACL and Security Group

Feature	NACL	Security Group
Supports	Allow and Deny Rules	Only Allow Rules
Stateful/Stateless	Stateless	Stateful
Association	Subnet	EC2 Instance
Rule Processing	Rules are evaluated in order	All rules are evaluated
Applied to	All instances in subnet	Only specified instances
Layer of Defense	Second Layer	First Layer

Public vs Private IP Address

Public IPs are used for internet communication, while private IPs are for internal network communication.